

Laboratory Record 5

Course :	Full Stack Web Development
Course Code:	23CM4121
Experiment no.:	5
Student Name:	R. Srivalli
Roll No.:	A24126552108

1. TITLE

Implement Modules in NodeJS

2. AIM

To implement and demonstrate **modules in Node.js** by creating a custom module containing reusable mathematical functions and importing it into another JavaScript file to perform arithmetic operations.

3. OBJECTIVE

1. To understand the concept of modules in Node.js.
 2. To create a custom module containing reusable functions.
 3. To export functions using module.exports.
 4. To import the custom module using the require() function.
 5. To perform arithmetic operations using functions from the imported module.
 6. To understand how modules improve code organization, reusability, and maintainability.
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4. THEORY

Node.js Modules are used to organize JavaScript code into separate, reusable files. A module can contain functions, variables, or objects that can be used in other JavaScript files.

Node.js commonly uses the CommonJS module system, where module.exports is used to export functions or values from a file, and require() is used to import them into another file.

In this program, a custom module named mathModule.js is created. It contains functions for addition, subtraction, multiplication, and division. These functions are exported using module.exports. The file scriptrec5.js imports the module using require("./mathModule") and calls the required functions to perform arithmetic operations.

Using modules makes programs well-organized, reusable, maintainable, and easier to manage.

5. ALGORITHM / PROCEDURE

1. Create a JavaScript file named `mathModule.js`.
 2. Define functions for addition, subtraction, multiplication, and division.
 3. Export these functions using `module.exports`.
 4. Create another JavaScript file named `scriptrec5.js`.
 5. Import the custom module using `require("./mathModule")`.
 6. Initialize two numbers, `a = 20` and `b = 5`.
 7. Call the arithmetic functions using the imported module.
 8. Display the calculated results using `console.log()`.
 9. Open the terminal in the `src` folder.
 10. Execute the program using: `node scriptrec5.js`
 11. Verify the addition, subtraction, multiplication, and division results.
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6. PROGRAM

A : `mathModule5.js`

```
// Experiment 5: Implement Modules in NodeJS
// User-defined module for mathematical operations

function add(a, b) {
    return a + b;
}

function subtract(a, b) {
    return a - b;
}

function multiply(a, b) {
    return a * b;
}

function divide(a, b) {
    if (b === 0) {
        return "Cannot divide by zero";
    }
}
```

```
    return a / b;
}

// Export the functions so they can be used in another file
module.exports = {
    add,
    subtract,
    multiply,
    divide
};
```

B : script.js

```
// Experiment 5: Implement Modules in NodeJS

// Import the user-defined mathematical module
const math = require("./mathModule5");

// Input values
const firstNumber = 20;
const secondNumber = 5;

// Display the results
console.log("----- NodeJS Module Demonstration -----");

console.log("First Number :", firstNumber);
console.log("Second Number :", secondNumber);

console.log("Addition      :", math.add(firstNumber, secondNumber));
console.log("Subtraction   :", math.subtract(firstNumber, secondNumber));
console.log("Multiplication :", math.multiply(firstNumber, secondNumber));
console.log("Division       :", math.divide(firstNumber, secondNumber));
```

7. CODE EXPLANATION

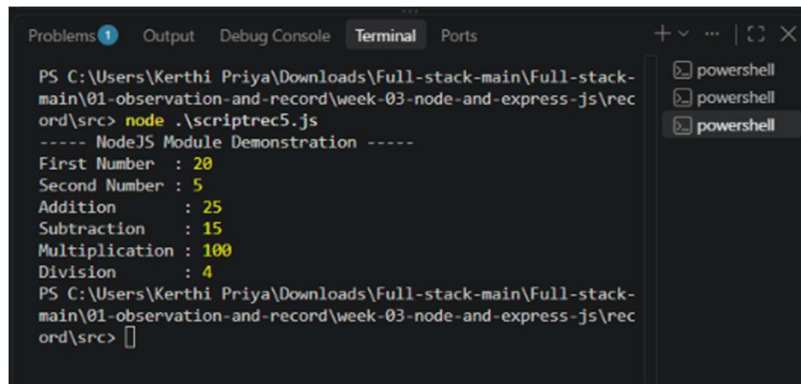
S.No.	Code / Component	Explanation
1	function add(a, b)	Defines a function to calculate the addition of two numbers.
2	function subtract(a, b)	Defines a function to calculate the difference between two numbers.
3	function multiply(a, b)	Defines a function to multiply two numbers.

4	function divide(a, b)	Defines a function to divide the first number by the second number.
5	return a + b	Returns the result of addition.
6	return a - b	Returns the result of subtraction.
7	return a * b	Returns the result of multiplication.
8	return a / b	Returns the result of division.
9	module.exports	Exports the arithmetic functions so they can be used in another JavaScript file.
10	require("./mathModule")	Imports the custom mathModule.js file into scriptrec5.js.
11	const a = 20	Stores the first input number as 20.
12	const b = 5	Stores the second input number as 5.
13	math.add(a, b)	Calls the add() function from the imported module.
14	math.subtract(a, b)	Calls the subtract() function from the imported module.
15	math.multiply(a, b)	Calls the multiply() function from the imported module.
16	math.divide(a, b)	Calls the divide() function from the imported module.
17	console.log()	Displays the input values and calculated results in the Node.js terminal.

8. TEST CASES AND EXECUTION DATA

Test Case 1 – Normal Program Execution

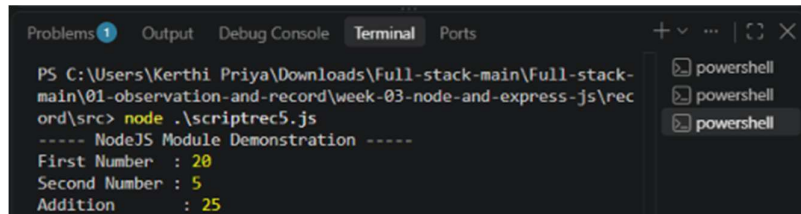
Input / Action	Expected Result	Status
Set a = 20 and b = 5, then run node scriptrec5.js	Program executes successfully and displays all arithmetic results	Pass



```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src> node .\scriptrec5.js
----- NodeJS Module Demonstration -----
First Number : 20
Second Number : 5
Addition : 25
Subtraction : 15
Multiplication : 100
Division : 4
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src>
```

Test Case 2 – Addition

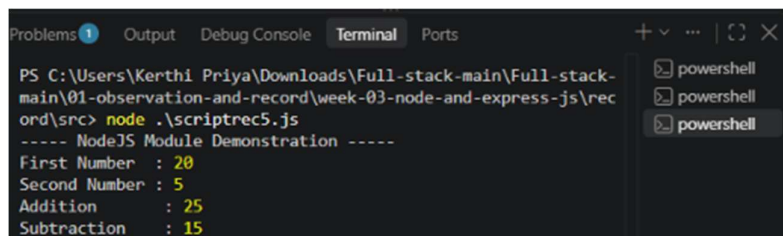
Input / Action	Expected Result	Status
Enter a = 20 and b = 5, then call the addition function	Addition result 25 is displayed	Pass



```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src> node .\scriptrec5.js
----- NodeJS Module Demonstration -----
First Number : 20
Second Number : 5
Addition : 25
```

Test Case 3 – Subtraction

Input / Action	Expected Result	Status
Enter a = 20 and b = 5, then call the subtraction function	Subtraction result 15 is displayed	Pass



```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src> node .\scriptrec5.js
----- NodeJS Module Demonstration -----
First Number : 20
Second Number : 5
Addition : 25
Subtraction : 15
```

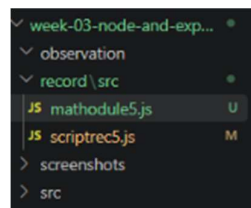
Test Case 4 – Negative Values

Input / Action	Expected Result	Status
Set a = -20 and b = 5, then run the program	Correct arithmetic results are displayed	Pass

```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src> node .\scriptrec5.js
----- NodeJS Module Demonstration -----
First Number : -20
Second Number : 5
Addition : -15
Subtraction : -25
Multiplication : -100
Division : -4
```

Test Case 5 – Missing Module File

Input / Action	Expected Result	Status
Remove or rename mathModule.js, then run scriptrec5.js	Program should handle the missing module without an error	Fail



```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src> node .\scriptrec5.js
node:internal/modules/cjs/loader:1568
  throw err;
  ^

Error: Cannot find module './mathModule5'
Require stack:
- C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src\scriptrec5.js
    at Module._resolveFilename (node:internal/modules/cjs/loader:1564:15)
    at wrapResolveFilename (node:internal/modules/cjs/loader:1118:27)
    at defaultResolveImplForCJSLoading (node:internal/modules/cjs/loader:1142:10)
    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1169:12)
    at Module._load (node:internal/modules/cjs/loader:1341:5)
    at wrapModuleload (node:internal/modules/cjs/loader:261:19)
    at Module.require (node:internal/modules/cjs/loader:1674:12)
    at require (node:internal/modules/helpers:157:16)
    at Object.<anonymous> (C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src\scriptrec5.js:4:14)
    at Module._compile (node:internal/modules/cjs/loader:1929:14) {
  code: 'MODULE_NOT_FOUND',
  requireStack: [
    'C:\\Users\\Kerthi Priya\\Downloads\\Full-stack-main\\Full-stack-main\\01-observation-and-record\\week-03-node-and-express-js\\record\\src\\scriptrec5.js'
  ]
}

Node.js v24.21.0
```

Test Case 6 – Division by Zero

Input / Action	Expected Result	Status
Set a = 20 and b = 0, then call the division function	Program should display a suitable division-by-zero error	pass

```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation
node-and-express-js\record\src>
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src> node .\scriptrec5.js
----- NodeJS Module Demonstration -----
First Number : -20
Second Number : 0
Addition : -20
Subtraction : -20
Multiplication : -0
Division : Cannot divide by zero
```

9. OUTPUT

Output 1 – Node.js Module Execution

```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-03-node-and-express-js\record\src> node .\scriptrec5.js
----- NodeJS Module Demonstration -----
First Number : 20
Second Number : 5
Addition : 25
Subtraction : 15
Multiplication : 100
Division : 4
```

Output 2 – Module File Structure

```
record\src
├── mathmodule5.js
├── scriptrec5.js
├── screenshots
└── src
```

10. RESULT

The Node.js module program was successfully implemented using a custom CommonJS module. The arithmetic functions were exported using `module.exports` and imported using `require()`. The program successfully performed addition, subtraction, multiplication, and division for the given values 20 and 5, producing the expected results.